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Determining supply chain effectiveness for Indian MSMEs: A structural equation modelling approach

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ABSTRACT

Industry 4.0 has gained recent attention due to the dynamic nature of businesses. Implementation of technologies of industry 4.0 offers numerous benefits to the business, and the supply chain is no exception to it due to the growing need to adopt new technologies, especially in MSMEs, as they face liquidity issues. This study is conducted to determine the adoption factors for supply chain finance in MSMEs, eventually enhancing supply chain effectiveness. This study has considered negotiation, collaboration, the external environment, digitizing trade, and the role of financial institutions. The study uses PLS Based Structural Equation Modelling to analyze the proposed conceptual model wherein the data was collected from various MSMEs in India. Results indicate that internal factors like negotiation, collaboration, and digitizing trade play an essential role in adopting supply chain effectiveness, and implementing supply chain finance leads to improved supply chain effectiveness.

1. Introduction

These days' the supply chain has become complex due to the growing size of the businesses, diverse product portfolios, and various geographic places to be served (Kamble, Gunasekaran, & Sharma, 2018). In Industry 4.0, supply chain management has gained importance due to the growing need to adopt new technologies. Supply chain operations have become dynamic due to competitive pressure and global network (Creazza et al., 2010; Zhang et al., 2003), and therefore, Supply chain (SC) management has become a challenging task for managers. SC management focuses on firms having the capacity to set and maintain long-term relationships with partner organizations (Fei & Yi-na, 2006). To date, the primary focus of supply chain managers is on optimizing the flow of goods. Many studies have considered the movement of goods in the supply chain (Bhatnagar & Teo, 2009), but one of the essential aspects overlooked in previous studies is the financial flow in the supply chain (Basu & Nair, 2012; Pfohl & Gomm, 2009). Any organization's supply chain impacts the amount of working capital required by the firm, which shows it also affects profitability. Micro, small and medium enterprises (MSMEs) are also looking for alternatives for short-term financing that can provide assistance for fixing credit issues and enhance financial performance. In addition to other financing methods, organizations with higher bargaining power are getting trade credit from suppliers, worsening the situation for members in the upstream supply chain. (Coulibaly et al., 2013). These factors have led to determining solutions that can optimize working capital. In times of industry 4.0, SCF has emerged as an effective tool that helps optimize financial flows at different organizational levels (Hofmann, 2005), and the solutions are executed by financial institutions (Chen & Hu, 2011).

In India, an enterprise is considered a micro-enterprise when the amount of investment in plant and machinery does not exceed one crore and the turnover is not exceeding the amount of five crores. Small enterprise has an investment of less than or equal to ten crores, and the amount of turnover does not exceed fifty crores. Medium enterprises have the investment in plant and machinery not exceeding fifty crores, and turnover does not exceed two hundred and fifty crores (MSME, 2020).

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It is expected that the recent global pandemic may create uncertainty for world economies. (Fernandes, 2020; Ivanov, 2020). Practitioners and researchers consider strengthening supply chain management to deal with such disruptions. (Jacobsen, 2020). In the context of India, the result of a survey shows that the earnings of MSMEs in India have been affected by 20–50% due to problems with liquidity (Tripathi, 2020). Earlier studies have reported that adopting Industry 4.0 is difficult for MSMEs due to the scarcity of funds for adopting digital technologies. (Erol et al., 2016; Müller & Voigt, 2016).

In India, MSME contributes 29% to the gross domestic product (GDP) and 49% to exports of the country, whereas the Government of India (GoI) plans to enhance this to 50% and 75%, respectively (PwC India, 2020a, 2020b). However, many MSME firms are excluded from formal lending ecosystems and do not have access to credit facilities for working capital requirements, which affects the rate of growth (PwC India, 2020a, 2020b). Despite GoI measures to extend support to MSMEs, this has remained a problem area; this may be due to the reluctant financial institution to lend to MSMEs due to high risk, although NPA rates of MSMEs are low compared to large firms. In India, firms require supply chain financing for inventory financing, post-shipment credit, WIP financing, and working capital financing. Supply chain financing is provided by commercial banks and other financial institutions in India and is obtained by MSMEs to minimize risk and costs. Traditional finance schemes take the collateral from the firm seeking finance which increases the risk. (Zhao et al., 2015). Latterly, SCF is introduced to minimize the transaction cost and reduce the default by firms. Lenders assess SMEs' creditworthiness and remain updated on buyer and seller information about transactions, orders, inventory, products, finances, and technology. SCF may optimize working capital and will eventually improve SCE. SCF has the potential to create long-term relationships with vendors by negotiating payment terms. Firms are adopting SCF to minimize risks, and the firm's supply chain performance can be enhanced without losing suppliers (Phillip, 2010).

Existing literature has provided two views about SCF. One relates to the financial benefits (More & Basu, 2013). The second perspective suggests that SCF provides monetary benefits and creates relationships among the parties involved in the supply chain (Gelsomino et al., 2016). Many recent articles have stated that SCF is a technique that can optimize working capital, but other factors are ignored. Therefore, more such studies are needed to determine the factors that affect SCF adoption in MSMEs. This study proposed a model focusing on the transaction cost approach; this implies that supply chain financing provides a solution that can minimize the transaction cost and provides a mechanism that helps optimize liquidity and enhances supply chain effectiveness. The current study focuses on various factors affecting SCF adoption, which are paramount in financial matters. Extant literature is not providing any encouraging results for the firms to adopt SCF, and therefore this study intends to fill this gap. This study is empirical and analyses the five factors that may affect SCF adoption in a firm.

This study aims to identify the factors affecting SCF adoption by MSMEs. The study will also include that adopting SCF improves the firm's liquidity position, eventually improving the supply chain effectiveness. Transaction cost is also less in comparison to traditional financing. This study makes a significant contribution by proposing a model leading to supply chain effectiveness among MSMEs. This study extends work conducted by Ali et al. (2019), as the external environment variable is also included as an additional construct. Ali et al. (2019) conducted their study on Pakistan's SMEs, and we wanted to test the model in the Indian scenario to examine whether those factors are valid for Indian SMEs. Indian SMEs are very different from those of developed nations, and technology adoption is not significant in Indian SMEs. External Environment was considered an additional construct as Indian SMEs are naturally cash crunch organizations that are less resilient to external shocks like the Covid-19 pandemic. Thus, we wanted to test the external environment's effect on Indian SMEs by extending their four-factor model to a five-factor model in the Indian scenario. This study is conducted considering industry 4.0, which suggests that the implementation of technologies will minimize the transaction cost if a firm employs SCF to manage SC effectively. Also, firms can seek financing from financial institutions without providing any collateral. Despite being such an important area for study, very few contributions are seen in SCF in India, claiming further study is required to enrich the literature.

Further sections are organized as follows: Section 2 reviews the related literature and develops the hypotheses for the study; section 3 includes the research methodology, including sample and measures used for collection and analysis of data; section 4 reports the analysis and results of the study; section 5 reports the conclusion, and section 7 offers the Implications and limitations of the study.

2. Literature review and hypotheses development

2.1. Factors affecting the adoption of SCF in MSMEs

A survey conducted by PwC India (2020b) states that buyers and suppliers view SCF as a tool for optimizing working capital and reducing the cash conversion cycle; this was considered a crucial benefit followed by strengthening the buyer-seller relationship. SCF will help the firms in running their business smoothly. Previously, the primary focus of SCM was on the physical flow of goods, and financial flow was neglected. However, recently it has started gaining importance as effective SCM optimizes the physical flow of goods but also the financial flow of the organization. Stemmler and Seuring (2003) were among the first few authors to introduce the concept of SCF.

The transaction cost is a major concern in real situations while obtaining funds from financial institutions. (Song et al., 2016). Many MSMEs find it difficult to obtain finance from financial institutions due to a lack of credit history and high transaction costs. (Song & Wang, 2013). In the case of MSMEs, it was observed that sometimes transaction cost is too high and exceeds the credit amount, due to which MSMEs do not find it attractive to take financing under traditional schemes. (Song et al., 2016). Therefore, this emerging tool, "SCF," may cater to the needs of MSMEs and minimize the risk of default, which will create a win-win model for both borrowers and financiers. Despite so many benefits, so many challenges affect SCF adoption in MSMEs. Therefore, it becomes imperative to analyze the adoption factors which play a vital role while implementing SCF. The study by Randall and Farris (2009) examines that the implementation of SCF in a firm increases the level of commitment and trust among the SC players. Lekkakos and Serrano (2016) found that SCF enhances suppliers' operational performance and optimizes the working capital.

In the study, five factors are considered to assess whether they affect the adoption of SCF in MSMEs. These factors are negotiation, collaboration, digitizing trade, external environment, and financial institutions. Also, it is analyzed whether these factors enhance supply chain effectiveness.

2.2. Hypotheses development

Negotiation is a term that suggests two people agreeing on a shared decision for a problem about which differences existed earlier (Carnevale & Isen, 1986). Liebl, Hartmann, & Feisel, 2016 stated that negotiation between buyer and supplier is essential in determining SCF adoption in a firm. SCF may create a win-win situation for buyer and seller by adopting an integrative strategy instead of a distributive strategy. In distributive strategy, one argues with another party aggressively for convincing to agree on a lower price or reduced delivery time (Walton & Mckersie, 1965). This strategy leads to a win-lose situation price (Walton & Mckersie, 1965). Adopting an integrative strategy requires trust and openness from other parties (Adair, 2003). Balance of power between buyer and seller is of paramount importance for the adoption of SCF. Therefore, hypothesis 1 states that:

H1. Negotiation between SMEs and suppliers has a positive relationship with SCF

Collaboration is an essential element while adopting SCF. Collaboration signifies trust and commitment among the SC players. Many researchers have identified trust as an influential antecedent of effective supply chain collaboration (Capaldo & Giannoccaro, 2015). Firms at all levels of the supply chain must work jointly to minimize SC threats and achieve a shared objective of forecasting and predicting. (Kamble et al., 2020). Collaboration among the SC partners creates bonding and creates an environment for planning jointly and sharing information on a real-time basis (Jüttner & Maklan,·2011), which is very much needed for overcoming the negative impacts of disasters (Altay et al., 2018). Collaboration among the SC players enhances the performance of SC (Kanwal & Rajput, 2016) and, eventually, better firm performance. It is widely accepted that collaboration in SC leads to better performance in SMEs (Fawcett et al., 2012). Vieira et al. (2015) also highlighted that interpersonal collaboration improves SC performance by reducing transaction costs. The benefits of SCF in industry 4.0 cannot be availed without a high level of integration among the supply chain partners, as all members in the supply chain have to adapt to the technology. Therefore, hypothesis 2 states that:

H2. Collaboration among SMEs and suppliers has a positive relationship with SCF

Banks or financial institutions provide financing to MSMEs, which suggests the role of financial institutions becomes a significant factor for a firm implementing SCF solutions. In India, many financial institutions grant loans to MSMEs, and therefore the attitude of top management is highly significant. Financial institutions which provide financing to MSMEs carry out more detailed discussions so that steps can be taken to minimize risks arising from asymmetric information and can offer services for integrated management of SC flow. Relationship among SC players is vital to measure for providing loans to MSMEs by the financial institutions. Generally, institutions also collect payments from the SC players to increase revenue (Tanrisever et al., 2012). Therefore, hypothesis 3 states that:

H3. Supporting role of financial institutions has a positive relationship with SCF

Implementation of Blockchain technology (BT) in the supply chain may change the way transactions are conducted (Kamble, Gunasekaran, & Himanshu, 2018). In the past also, technology-driven disruptions have improved the efficiency of businesses. (Craighead et al., 2007). The supply chain is no exception, as technology has also impacted this stream. Nowadays, looking at businesses catering to diverse locations, the supply chain has gained importance and has become one of the most crucial elements. With the usage of digital technologies in the supply chain, firms can take a competitive advantage (Gunasekaran et al., 2008), which will help them make decisions. Decisions like predicting demand, production planning, and syncing orders are primary challenges in the uncertain environment (Pereira, 2009). Therefore, moving towards digitalization is of paramount importance to minimize errors in such important decisions. Firms may adopt Big Data Analytics for timely information processing, thereby making suitable decisions (Kamble & Gunasekaran, 2020). Technologies like the Internet of things (IoT), BT, a digital twin, can improve supply chain resilience due to their connectivity, accuracy, and transparency (Kamble et al., 2020[b]). SMEs are adopting a digital system to place the orders, due to which it is becoming easier to check the status of placed orders among the sc players (Fairchild, 2005). Adopting technology reduces the cost of paperwork (Perego & Salgaro, 2010), but implementation costs act as a challenge. There is a considerable cost in implementation for developing the infrastructure for industry 4.0 (Kamble, Gunasekaran, & Sharma, 2018). Digitizing trade offers flexibility to the SC partners, and by adopting an electronic platform, MSMEs can enhance transparency. As per a study by Caniato et al. (2016), firms that digitize their trade activities tend to adopt advanced financing solutions. Therefore, hypothesis 4 states that:

H4. Digitization of trade has a positive relationship with SCF

Implementation of SCF has to be conducted under the umbrella of macro-economic variables. Macroeconomic variables inevitably affect businesses, and therefore this is one such factor that will affect the decision to adopt the SCF in an MSME firm. Macroeconomic conditions of an economy affect a firm's decisions like investing, financing, and promotion decisions. As financial institutions or banks do, the financing, laws, regulations, monetary policy, and other relevant frameworks also affect SCF decisions (Zhang, 2015). In India, banks are governed by the Reserve Bank of India, so the governing body's rules also influence the firm's businesses. Credit allocation is one such decision that affects MSMEs predominantly. MSMEs will not grow and develop unless institutions meet their credit requirement cost-effectively and efficiently. External Environments are also considered as the Covid-19 pandemic brought in an unpredictable demand, and the use of technology was enhanced as all other physical activities were shut down. Furthermore, Indian MSMEs have a severe talent shortage as they rely on daily wage employees rather than full-time employees, so it became evident to test the impact of the external environment on the supply chain finance of MSMEs. Therefore, the role of the external environment is an important factor that may affect the decision to adopt SCF in a firm. Therefore, hypothesis 5 states:

H5. Role of External Environment has a positive relationship with SCF

In order to minimize the threats to the survival of the business, firms shall do better planning. Therefore, SC development may benefit the firm by meeting financial and operational objectives. Gunasekaran, Patel, and McGaughey (2004) also highlighted that achieving supply chain effectiveness shows the firm level of success in meeting operational and financial goals. Various factors of supply chain finance improve SC responsiveness, thereby leading to supply chain effectiveness. A supply chain is considered adequate if the handling and shipping costs are minimized; this reduces the price of products or services, thereby benefiting the firm and other stakeholders. Jing and Seidmann (2014) also highlighted that trade financing is better than bank financing. Wuttke et al. (2016) highlighted that SCF's implementation improves SC's performance by extending payment terms for buyers and optimizing working capital for the supplier to ensure liquidity in the business. Therefore, hypothesis 6 states:

H6. SCF has a positive relationship with SCE (Fig. 1)

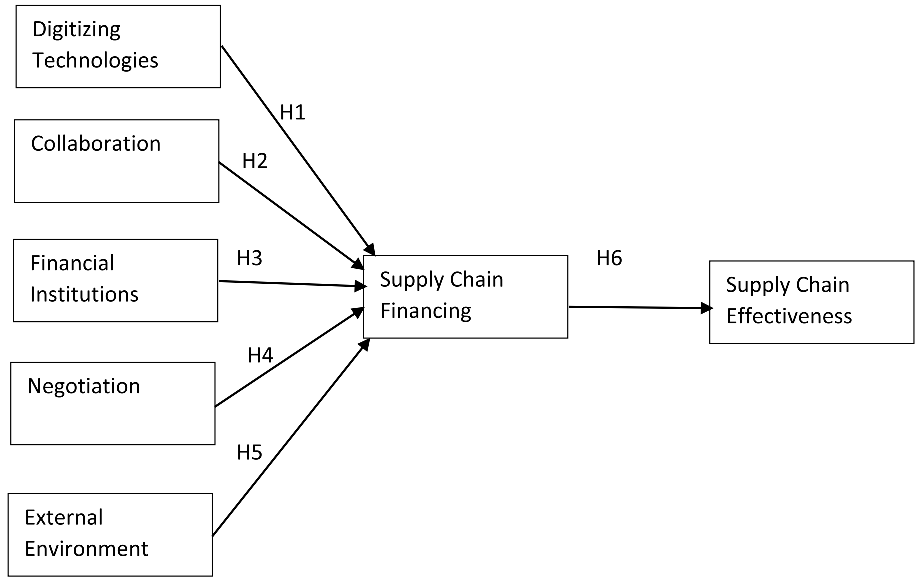

Fig. 1. Proposed conceptual model for the study.

3. Research methodology

3.1. Sample

Supply chain financing in Micro and Small Medium Enterprises (MSMEs) is increasing day by day in India. The participants for the current study were selected from various MSMEs (especially Micro and Small Organizations) located in four states in India, namely Indore in Madhya Pradesh, Samba, and Bari-Brahmana in Jammu and Kashmir; Bathinda and Gurdaspur in Punjab; and Surat in Gujarat. The respondents included people from MSMEs, including owners, Executives in the supply chain, and Managers in the supply chain of an MSME. The locations were selected because they are industrial hubs in the state and based on the authors' locational convenience.

.The selection of MSMEs was based on the fact that they are more prone to risks and crisis-like situations. The Covid-19 pandemic has had a significant impact on MSMEs in India, and given the fact that more than 99 percent of Indian businesses are from Micro organizations, it makes a significant impact on the Indian Economy (Sahoo & Ashwani, 2020). Indian MSMEs play a crucial role in overall economic development as more than 35 percent of contribution to overall GDP comes from them (Aneja & Ahuja, 2020).

According to the Micro, Small, and Medium Enterprises Development (MSMED) Act, 2006, Indian organizations falls into three categories i.e., micro, small, and medium. Every country has a different criterion while classifying MSMEs and defining them as some define them based on the number of employees engaged, annual sales and turnover, or investment. In India, as per the MSMED Act, 2006, MSMEs are defined based on investment by enterprises in plant and machinery. Enterprises in Manufacturing and Services having investments up to INR 1 crores fall under the category of Micro Enterprises, Enterprises having investments more than one crore but less than ten crores fall under the category of Small Enterprises, and Enterprises having investments more than ten crores and less than twenty crores fall under the category of medium enterprises. To collect the data, investment and the number of employees in MSMEs were considered essential criteria. Enterprises having less than 200 employees were approached for data collection purposes.

The participants were approached from January 2020 to October 2020. Initially, until March 2020, the participants were contacted by phone, and after taking approval to participate in the study, they were approached, and questionnaires in pen and paper form were filled out. Owing to the lockdown restrictions, the data collection after March 2020 was done digitally, and a google form similar to a pen and paper questionnaire was created and was floated to the participants after taking their due approval over the telephone. The participants were briefed about the purpose of the study, and those participants who were not comfortable with English were connected through Zoom or WhatsApp video call, and the questions were translated to them into Hindi, and their responses were recorded accordingly.

Furthermore, to ensure the reliability of the questions in the questionnaire, the questionnaire was checked by three area experts (two professors from Management Institutes and one Industry person). The data collection process took ten months to complete because of the Covid-19 crisis, and 274 filled responses were received.

3.2. Measures

The measurement scales for the present study were taken from various available literature published in the area to ensure the validity and reliability of the instrument used. The scale for the dependent variable Supply Chain Effectiveness (SCE) was adapted from Fugate et al. (2009), and the items were modified as per the objective of the current study; the scale for supply chain finance (SCF) was adapted from Ali et al. (2019) with five items; the scale for digitizing technologies (DT) was adopted from Choi (2013) using three-item scale; the scale for collaboration was taken from Simatupang and Sridharan (2005) using a four-item scale; the scale for Financial institutions was adapted from Zhang (2015) with a three-item scale; the negotiation scale was adapted using three items from Janda and Seshadri (2001); and the scale for External Environment was adapted using a three-item scale from Zhang (2015).

4. Data analysis and results

The data analysis for the study was done using SPSS version 24 and SMART PLS version 3.3.2, the method of "*partial least squares*

Table 1
Reliability and validity of Constructs.

Constructs	Cronbach's Alpha	Composite Reliability	Average Variance Extracted (AVE)
Collaboration	0.892	0.923	0.752
External Environment	0.880	0.841	0.647
Digitizing Technologies	0.903	0.939	0.838
Financing Institutions	0.834	0.900	0.749
Negotiation	0.777	0.813	0.601
Supply Chain Effectiveness	0.813	0.864	0.614
Supply Chain Finance	0.825	0.879	0.596

Table 3
Heterotrait-Monotrait (HTMT) ratio of correlations.

Constructs	CO	DT	EE	FI	NE	SCE	SCF
Collaboration							
External Environment	0.340						
Digitizing Technologies	0.266	0.157					
Financing Institutions	0.306	0.137	0.311				
Negotiation	0.116	0.198	0.174	0.328			
Supply Chain Effectiveness	0.170	0.100	0.152	0.474	0.290		
Supply Chain Finance	0.316	0.116	0.479	0.375	0.307	0.530	

structural equation modeling" (PLS-SEM). The objective of the current study is to explore the factors leading toward Supply Chain Effectiveness and is an extension of the existing study by Ali et al. (2019); thus, the use of PLS-SEM is done as suggested by Hair et al. (2011). Before running the data for measurement and structural analysis, we checked the data for multi-collinearity and missing data values. No such concerns with the data were found. The collinearity (VIF) values (Annexure-I) were less than the cut-off values of 3.5, indicating ideal values (Hair et al., 2019). Thus, we proceeded with Measurement model analysis to examine the reliability and validity of the constructs. In Table 1, scale reliability and validity through the measurement model is performed and next step of structural model analysis and hypothesis testing is conducted.

4.1. Measurement model analysis

Being a standard procedure for SEM analysis, the measurement model analysis was performed to check scale reliability and validity. The constructs used in the study attained Cronbach's alpha value above the recommended cut-off value of 0.7 (Fornell & Larcker, 1981). The construct for negotiation depicted the lowest Cronbach's alpha value at 0.777, which is above the minimum threshold value (See Table 2). The items thus indicate high internal consistency and reproducibility of the scale (Fornell & Larcker, 1981).

To test the Convergent validity of the scale, the Average Variance Extract (AVE) measures were employed. The results of all constructs depicted a value of more than 0.5, indicating satisfactory convergent validity among constructs (Fornell & Larcker, 1981). The outcome of AVE depicted values from 0.596 to 0.838 (See Table 2).

Using the Fornell and Larcker Criteria (1981), discriminant validity was tested by comparing the squared root values of AVE with the inter-constructs correlations. As depicted in Table 3, the square root values highlighted in bold are more than the inter-construct correlation values. Thus implying that the measurement model used has discriminant validity and the constructs used in the study differ from other study constructs. However, the use of discriminant validity has been a subject of debate in multiple studies recently, where some researchers have criticized the criteria indicating a lack of reliability of the criteria in variance-based SEM studies (Voorhees et al., 2016; Shiu et al., 2011). Thus, the Heterotrait and Monotrait ratio of correlations (HTMT) method was further applied to examine the discriminant validity of the measurement model (Henseler et al., 2015). HTMT is a new advancement in PLS to determine the discriminant validity among the variables (Carrión et al., 2016). The results for all seven constructs met the cut-off criteria of 0.85 (Henseler et al., 2015). Table 4 provide the correlation among the variables and all constructs had low correlation, thus meeting the criteria of HTMT.

4.2. Structural model analysis

After confirming the measurement model, the structural model analysis was performed. For the structural model analysis, the SEM procedure of PLS is used to analyze the goodness of fit index, R square, and hypotheses testing (Path coefficients). To check the path coefficients bootstrapping procedure with 5000 sub-samples was performed.

The model's overall Goodness of Fit (GoF) is 0.517, obtained from the outlines (Alolah et al., 2014) using the Equation $\sqrt{\overline{AvE}*\overline{R^2}}$. Using the equation, the average AVE scores of seven constructs an average R square of two endogenous variables (SCE and SCF) are taken into consideration. Considering the criteria of GoF values of small at 0.1, medium at 0.25, and large at 0.36 (Esfandiar et al., 2019), we can say that the current model has a good model fit. Furthermore, the Model Fit Indices (See Table 5) highlight good model fit using Standardized Root Means Square (SRMR), indicating a value less than the cut-off limit. Similarly, Normed Fit Index (NFI) also indicated a good model fit with a value of 0.956. Furthermore, the rms Theta value indicated a good model fit with a value of 0.089.

According to R Squared coefficients, the constructs modeled for the current study explained a moderate variance, with 40.7 percent variance reported by Supply Chain Financing and 36.7 percent variance reported by Supply Chain Effectiveness. The R square variance lies at a satisfactory level of more than 0.10, as suggested by Hair et al. (2019). Possible reasons for getting moderate R square values can be the presence of factors not covered in the study due to their scope.

Table 2
Discriminant Validity using Fornell-Larcker Criteria.

Constructs	CO	EE	DT	FI	NE	SCE	SCF
Collaboration	**0.867**						
External Environment	0.243	**0.804**					
Digitizing Technologies	0.273	−0.186	**0.915**				
Financing Institutions	0.273	−0.146	0.280	**0.866**			
Negotiation	−0.028	−0.185	0.220	0.335	**0.775**		
Supply Chain Effectiveness	0.111	−0.081	0.157	0.423	0.344	**0.784**	
Supply Chain Finance	0.288	−0.089	0.420	0.321	0.368	0.517	**0.772**

Note: Values in Bold indicate Squared values of AVE Scores.

Table 4
Model fit indices.

Model Fit Indices	Saturated Model	Cut-Off Values
SRMR	0.063	<0.10 (Hu & Bentler, 1999)
Chi-Square	1609.713	
NFI	0.956	>0.90 (Lohmöller, 1989)
rms Theta	0.089	<0.12 (Henseler et al., 2014)

Table 5
R square values.

Constructs	R Square	R Square Adjusted
Supply Chain Effectiveness	0.367	0.364
Supply Chain Finance	0.407	0.394

After gaining confidence in the validity of the structural model, the path coefficients and hypotheses testing was performed. The results of PLS-SEM indicated solid acceptance to all the proposed hypotheses except H3 (Financial Institutions - > Supply Chain Finance: β = 0.139, t = 1.478) and H5 (External Environment - > Supply Chain Finance: β = 0.794, t = 0.261). In particular, the other direct effects indicated significant path coefficient results with supply chain finance indicating a strong positive impact on supply chain effectiveness (β = 0.00, t = 10.805). At the same time, collaboration, an indicator of Supply Chain Finance (β = 0.001, t = 3.407), indicated a positive relationship. Furthermore, Digitizing Technologies as an Indicator of Supply Chain Finance (β = 0.00, t = 4.742) indicated a positive and significant relationship. Lastly, Negotiation, an Indicator of Supply Chain Finance, also depicted a positive and significant relationship with (β = 0.00, t = 5.196) values. Table 6 provides detailed results of the hypotheses testing.

Table 7 provides factor loading, which is considered absolutely important in forming the formative construct when it is ≥ 0.50 and statistically significant (Hair et al., 2020).

5. Conclusion

This study attempts to determine the factors leading to SCF's adoption and supply chain finance's adoption to enhance supply chain effectiveness in MSMEs of India. This study shows that SCF not only considers financial factors, but practical implementation of SCF can be done when it is supplemented by negotiation, collaboration, and digitizing trade. Results indicate that firms should focus not only on the financial variables of the supply chain but also on negotiation, collaboration, and the extent of digitization in the firm. Digitization of trade is a crucial process that can help the Indian MSMEs grow by offering many benefits like sharing costs and reducing transportation costs. The findings for digitizing trade are also in line with Ali et al. (2018). Negotiation for MSMEs is a very prominent factor in Supply Chain Finance to ensure effectiveness in their supply chain as most MSMEs are cash crunch, and poor negotiation power in MSMEs can put pressure on their Return on Investment. The findings for negotiation as a factor of supply chain finance are similar to Thakkar et al. (2009). Collaboration is also a very prominent indicator of the supply chain effectiveness of a firm as most MSMEs have scalability issues which can affect the overall performance of the Indian MSMEs. Collaboration among MSMEs for the supply chain can overcome the financial and innovation constraints among Indian MSMEs. The findings are also in line with the findings of (Zaridis, Vlachos, & Bourlakis, 2021). The external environment and role of the financial institutions are not significant factors affecting SCF adoption in the Indian context. A possible reason for this is that a supply chain is an emerging tool in India, and therefore all internal factors are significant as SCF is at the developing stage, so firms need to create an internal environment at the initial stage. Negotiation, collaboration, and digitizing trade are significant adoption factors. As of now, there is a need to develop buyer-supplier long-term relationships, inter-firm collaboration, and adapt to digital needs. This study also proves that SCF leads to SCE. This is consistent with the study of Wuttke et al. (2016), wherein it was highlighted that implementing SCF improves the performance of the supply chain by giving a longer duration for payment and suppliers; it optimizes the working capital. This suggests that various factors of SCF increase the responsiveness of SC, which enhances SC effectiveness.

6. Implications and limitations

This study will contribute to the existing literature on supply chain finance and supply chain effectiveness. In the context of MSMEs in India, the role of negotiation, collaboration, and digitizing trade are significant factors impacting supply chain effectiveness. In the current study, both internal and external factors are considered, and the impact of these is assessed on supply chain effectiveness. External Environment has also been added as one factor that has become insignificant. This also presents that supply chain finance is an emerging tool that may minimize SC players' default.

From the viewpoint of managers, this study will provide a better understanding of the benefits, potential, and drawbacks of the already implemented solution, which will add knowledge to managers and lead to better decision-making. This study will be of use to the managers of MSMEs. This will also provide an understanding of the SCF process in optimizing the working capital and minimizing the liquidity issues. Also, SCF helps improve the players' relations in the supply chain. Therefore, managers can take advantage of SCF to maintain long-term relations among the SC players and improve the financial flow of goods. This study will also help the MSMEs determine the factors affecting the supply chain effectiveness and the necessary steps that firm managers can take to decide on specific processes to be improved or new processes to be adopted. In addition, SMEs owners can obtain financing to meet working capital requirements and minimize the credit risk.

Table 6
Hypotheses testing.

Hypothesized Path Models	OS	SM	SD	T-Stat	P Values (β)	Result
Collaboration - > Supply Chain Finance (H2)	0.201	0.206	0.059	3.407	**0.001**	Accepted
External Environment - > Supply Chain Finance (H5)	−0.022	−0.007	0.084	0.261	**0.794**	Rejected
Digitizing Technologies - > Supply Chain Finance (H4)	0.274	0.270	0.058	4.742	**0.000**	Accepted
Financing Institutions - > Supply Chain Finance (H3)	0.093	0.086	0.063	1.478	**0.139**	Rejected
Negotiation - > Supply Chain Finance (H1)	0.278	0.292	0.053	5.196	**0.000**	Accepted
Supply Chain Finance - > Supply Chain Effectiveness (H6)	0.517	0.523	0.048	10.805	**0.000**	Accepted

Table 7
Factor loading table.

Items	Collaboration	Digitizing Tech_	External Environment	Financing Institutions	Negotiation	Supply Chain Effectiveness_	Supply Chain Financing
CO1	**0.705**						
CO2	**0.935**						
CO3	**0.886**						
CO4	**0.922**						
DT1		**0.973**					
DT2		**0.800**					
DT3		**0.596**					
EE1			**0.948**				
EE2			**0.894**				
EE3			**0.902**				
FI1				**0.862**			
FI2				**0.851**			
FI3				**0.883**			
NE1					**0.676**		
NE2					**0.631**		
NE3					**0.973**		
SCE1						**0.835**	
SCE2						**0.789**	
SCE3						**0.811**	
SCE4						**0.693**	
SCF1							**0.890**
SCF2							**0.662**
SCF3							**0.877**
SCF4							**0.717**
SCF5							**0.682**

One of the enablers to supply chain finance is the digitalization of trade. MSMEs managers must contemplate digitizing trade as implementing SCF would become hassle-free. Due to a lack of digitalization, the firm may be forced to adopt the traditional mechanism, costing higher. This new method of financing may help minimize the default, and this will go on a long way toward improving relations between financial institutions with borrowers. Considering the potential benefits of SCF, adoption of SCF may increase among the MSMEs. MSMEs may consider and incorporate the factors like negotiation, collaboration, and digitizing trade that impact the adoption of SCF in a firm. Collaboration among the departments plays a vital role which affecting the implementation of SCF solutions. The results show that SCF acts as a success factor for improving SC effectiveness. This also provides a better understanding of the factors affecting SCF among SMEs managers.

This study is conducted on MSMEs in India, so there is a scope of the study by extending the research to other firms for generalization. Further, a comparison can be drawn between large-scale organizations and MSMEs to see the differences in the adoption factors affecting SCF. In the future, other factors like training and knowledge about SCF to employees, support from top management, integration of supply chain in the firm can be included to assess the impact of these on supply chain effectiveness. As per Randall and Farris (2009), the SCF approach leads to increased profitability, so there is further scope to see the impact of SCF on the firm's performance in the Indian context. Also, the impact of mediators and moderators can be tested on the model. Furthermore, other related SCF factor aspects, like Supply chain information, supply chain performance, and supply chain agility, can be tested in future studies to extend the current model. Future researchers can also examine the relationship of agency theory with supply chain financing.

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ANNEXURE – I: VIF Values indicating Multi-Collinearity among Items

Items	VIF
CO1	1.659
CO2	2.432
CO3	2.078
CO4	2.259
EE1	2.116
EE2	2.998
EE3	2.595
DT1	3.384
DT2	3.286
DT3	2.501
FI1	1.701
FI2	2.097
FI3	2.234
NE1	1.746
NE2	1.739
NE3	1.447
SCE1	1.347
SCE2	1.932
SCE3	2.071
SCE4	1.872
SCF1	2.666
SCF2	1.570
SCF3	3.454
SCF4	1.508
SCF5	1.398

ANNEXURE – II (Questionnaire: Items for Survey Scale)

Questionnaire Items (1 = “Strongly Disagree” to 5 = “Strongly Agree”)

Supply Chain Finance (SCF) (Adapted from Ali et al., 2019).

1. Supply chain finance is seen as a risk prevention system in our organization.

2. The capital flow coordination in the organization's supply chain is enhanced with Supply chain financing.
3. A high level of Supply chain efficiency comes with supply chain financing.
4. A high-risk prevention capability of an organization is enhanced with supply chain finance.
5. A high degree of technology is required for supply chain finance applications.

Supply Chain Effectiveness (SCE) (Adapted from Fugate et al., 2009).

1. Supply chain financing helps in reducing Transportation Cost
2. Supply chain financing helps in reducing warehousing Costs
3. Supply chain financing helps in reducing Inventory Cost
4. Supply chain financing helps in reducing Logistics Cost

Digitizing Technologies (DT) (Adapted from Choi, 2013).

1 The use of Digital technologies offers new ways of engaging buyers and suppliers
2 To trade with Buyers and Suppliers, training in digital technologies is required.
3 Relationship with buyers and suppliers is essential for the future of digital trading.

Collaboration (CO) (Adapted from Simatupang & Sridharan, 2005).

1. Our organization constantly shares Demand Forecast with suppliers
2. Our organization constantly shares Inventory Policy with Suppliers
3. Our organization constantly shares Price Changes with Suppliers
4. Our organization constantly shares any Supply Disruption with Suppliers

Financial Institutions (FI) (Adapted from Zhang, 2015).

1. A supportive attitude is shown by financial institutions while applying supply chain finance.
2. The Commercial banks have a perfect structure for implementing supply chain finance.
3. When an organization applies supply chain finance, risk prevention is better.

Negotiation (NE) (Adapted from Janda & Seshadri, 2001).

1. We as an organization seek a Win-Win opportunity
2. We as an organization spend time on the process of negotiation
3. We as an organization reach a mutual consensus on the problem situation

External Environment (EE) (Adapted from Zhang, 2015).

1. The macro-economic condition of an organization has a significant impact on Supply chain Finance
2. Laws and regulations in a country affect the choice of supply chain finance
3. The development of Supply chain finance technology helps in implementing supply chain finance.

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